MILITARY SPECIFICATION
DIPHENYLPHTHALATE
(TECHNICAL)

1. SCOPE

1.1 This specification covers the technical grade of diphenylphthalate for use in the manufacture of propellants.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATION

MILITARY

MIL-P-3324 - Paraffin, wax (refined, for histological pharmaceutical, and dental use)

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED FREIGHT CLASSIFICATION - Ratings, Rules, and Regulations.

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(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue, New York 16, New York.)

3. REQUIREMENTS

3.1 Material. - Diphenylphthalate shall be crystalline loose form or solid form.
3.2 Assay - When tested as indicated 4.4.1, ester content shall be no less than 99.0 percent.
3.3 Acidity (as phthalic acid). - Acidity shall not exceed 0.05 percent.
3.4 Ash content. - When tested as specified in 4.4.3, the ash content shall not be greater than 0.20 percent.
3.5 Loss in weight. - The loss in weight at 105°C. shall not be more than 0.5 percent.
3.6 Setting point. - Setting point shall be 69° to 70°C.
3.7 Workmanship. - Diphenylphthalate shall be in accordance with high grade commercial practice covering the manufacture of this material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provision. - Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.
4.2 Lot
4.2.1 Batch process. - A lot shall consist of the material produced from no more than one manufacturing batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.
4.2.2 Continuous process. - A lot shall consist of the material produced in no more than one work shift with no change in process or materials.
4.3. Sampling. - A 1/4 pound representative specimen shall be removed from each of five containers taken at random from each lot. The specimen shall be placed in a clean, dry container, thoroughly blended and labeled to identify the container and lot represented.
4.4 Test methods. - Distilled water and analytical reagent grade chemicals shall be used through out tests. Where applicable a blank determination shall be run and corrections applied where significant.
4.4.1 Assay. - Weigh 0.5 gram of the sample to the nearest .1mg. into adry 250 ml. Erlenmeyer flask with a ground glass joint and add 20 ml. of benzene. Dissolve by warming gently on an electric hot plate. Remove from the hot plate and while the solution is still warm, add 125 ml. of an approximately 0.5N solution of potassium hydroxide in anhydrous ethyl alcohol (prepared by dissolving 28 grams of potassium hydroxide pellets in 1 liter of anhydrous ethyl alcohol and filtering). Reflux at approximately 60°C for 2 hours, using a water condenser. Stopper, and cool to room temperature. Add 75 ml. of

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absolute ether and allow to stand 30 to 60 minutes. Filter through a tared sintered glass crucible (medium) and police the flask and wash the precipitate 6 times with 10 ml. portion of a mixture of equal parts of anhydrous ether and anhydrous alcohol. Do not allow the crucible to become completely dry during the filtering and washings. Heat the crucible for a few minutes at 80°C. then heat at 150°C. for 3 hours. Weigh as dipotassium phthalate. Calculate the percentage diphenylphthalate as follows:

$$\text{Percent diphenylphthalate} = \frac{131.36P}{W}$$

where:

P = weight of dipotassium phthalate
W = weight of sample

4.4.2 Acidity. - Measure 100 ml. of chloroform into a flask, add 3 drops of phenolphthalein indicator and titrate the chloroform to a faint pink color with 0.1 N sodium hydroxide solution. Add an accurately weighed 15 gm. portion of the sample to the chloroform solution. Titrate the mixture with 0.1 N sodium hydroxide. Calculate the percent acidity as phthalic acid as follows:

$$\text{Percent of phthalic acid} = \frac{8.3 \times V \times N}{W}$$

where:

V = volume of sodium hydroxide solution
N = normality of sodium hydroxide solution
W = weight of sample

4.4.3 Ash. - Weigh 1 gram of diphenylphthalate and approximately one gram of paraffin conforming to MIL-P-3324, and place in a 100 ml. tared procelain ash dish. Place the dish on a hot plate. Evaporate until nearly all the material has volatilized. Place the dish in 650° muffle for 30 minutes, cool and weigh. Run blank for this determination. Calculate percent of ash as follows:

$$\text{Percent ash} = \frac{\text{Wt. of ash}}{\text{Wt. of sample}} \times 100$$

4.4.4 Loss in weight. - Weigh 3 to 5 grams of sample, place in a tared shallow dish and dry to constant weight at 105°C. Calculate the loss as follows:

$$\text{Percent loss at } 105^{\circ}\text{C} = \frac{\text{Loss in wt.}}{\text{Wt. of sample}} \times 100$$

4.4.5 Setting point. -Dry 40 to 50 gm. of the sample for 4 hours at 40°C. Transfer to the inner tube of the solidification point apparatus shown on figure 1 and melt. Place the tube in the apparatus and adjust the standardized thermometer so that the bulb is in the center of the liquid with a side thermometer in position for the emergent stem correction. Stir the molten material vigorously by means of the hand stirrer and carefully note

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the point where the temperature begins to rise when solidification begins. Record the temperature every 15 seconds until the maximum reading is obtained. Correct the maximum temperature for emergent stem by adding the value calculated from the formula:

$$N(T - t) \times 0.000159$$

where:

N = degrees in the exposed mercury column
T = the uncorrected solidification point
t = average temperature of the exposed mercury column determined by means of a second thermometer suspended so that its bulb is in the mid point of the exposed mercury column
0.000159 = the coefficient of expansion of mercury in glass

Record the corrected reading as the solidification point of the sample.

4.4.6 Packing and marking. - It shall be ascertained that the packing and marking complies with the requirements of Section 5 of this specification.

4.4.7 Rejection. - The lot shall be rejected if the sample fails to comply with any of the requirements specified, whether tested by the contractor or check tested by the Government.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C. - The diphenylphthalate shall be packed in accordance with the manufacturer's standard commercial practice in such manner that will provide protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers used shall conform with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2 Marking. - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with standard MIL-STD-129.

6. NOTES

6.1 Intended use. - Diphenylphthalate is intended for use as deterrent coating in smokeless powder.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

Notice. -When Government drawings, specification, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way

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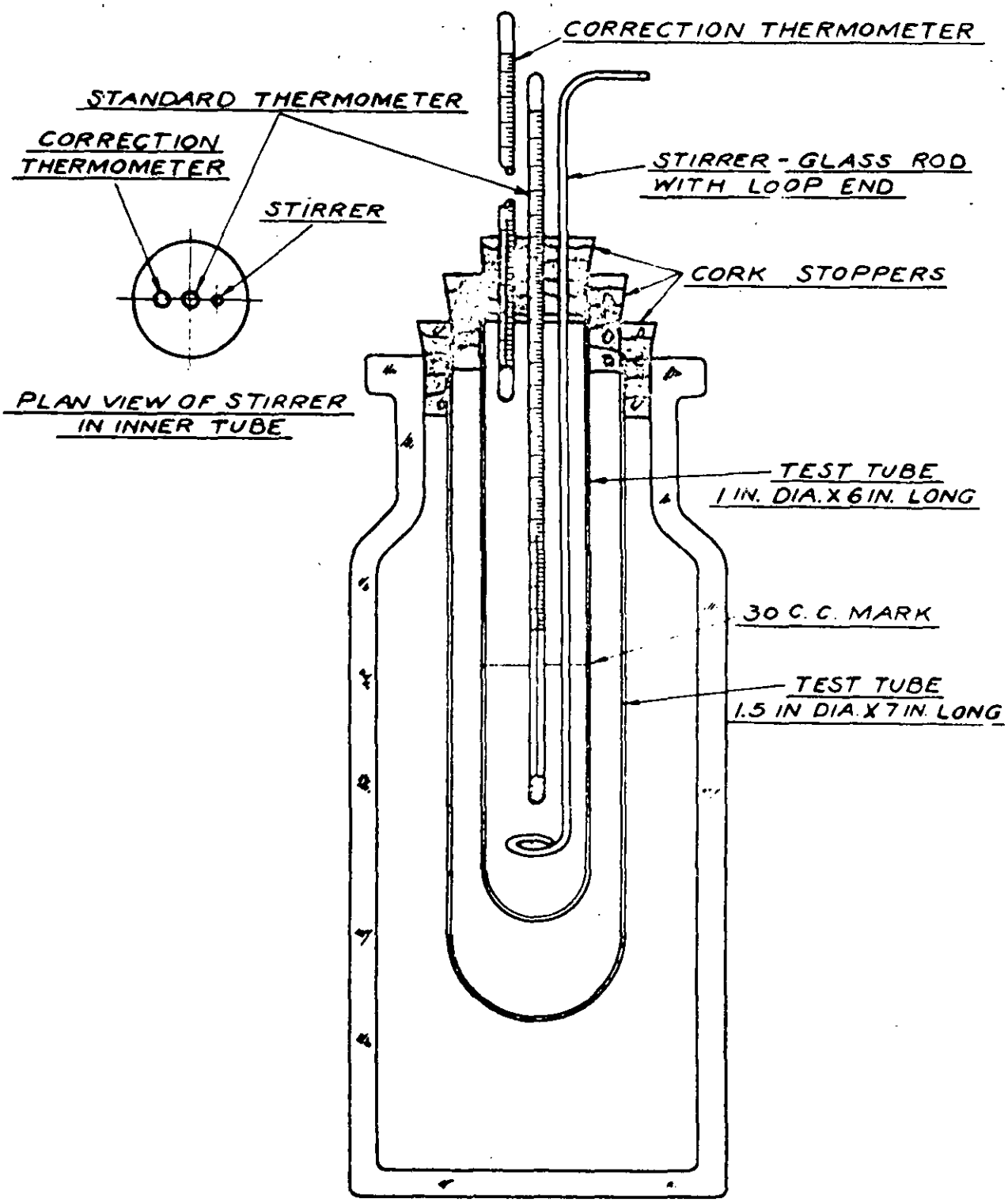
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APPARATUS FC DETERMINING SOLIDIF. ATION OR SETTING POINT



LARGE MOUTH LITER BOTTLE 1000 C.C. CAPACITY

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FIG. 1

GPO 868896